Assignment 2

1 Core Strategy

The most common way to predict ratings is Regression . However, this method has a ceiling because rounding loses information at the boundaries (e.g., is a 3.5 a ”3” or a ”4”?).

Instead, this solution treats the problem as a **5-Class Classification** task. This allows the model to:

- Directly optimize for the exact integer rating.
- Learn that some mistakes are costlier than others (e.g., confusing a 4 with a 5 vs. confusing a 1 with a 5).
- Output a probability distribution (e.g., 80% confident it’s a ”4”) rather than just a single number.

2 Neural Feature engineering using autorec

The first stage uses **AutoRec**, a specialized Neural Network (Autoencoder) designed for recommendation systems.

2.1 How it Works

The model takes a user’s known ratings, compresses them into a bottleneck layer to capture the essence of their taste, and then attempts to reconstruct the full list of ratings, including the ones the user hasn’t seen yet.

2.2 Two Perspectives

We train two different versions to capture the data from every angle:

- **Item-Based (I-AutoRec):** Analyzes which types of users usually enjoy a specific movie. This is the stronger model because movies generally have more consistent data than individual users.
- **User-Based (U-AutoRec):** Analyzes which movies users with similar tastes have enjoyed.

We blend these two to capture maximum information.

3 Ensemble Boosting

The raw scores from the Neural Networks are fed into an ensemble of three Gradient Boosting algorithms: **LightGBM, XGBoost, and CatBoost**.

3.1 Feature Engineering

Along with the scores from the previous step, we provide the ensemble with contextual features:

- **User Bias**
- **Item Popularity**
- **Model Disagreement**

Each boosting model has a different personality in how it grows decision trees. By averaging the probabilities from all three, we cancel out individual errors and produce a much more stable and accurate final prediction.

4 Why This Approach Wins

1. **Non-Linearity:** Unlike simple matrix factorization (SVD), the Neural Networks can find complex and not so obvious patterns in movie tastes.
2. **Bias Correction:** adjusts for relative rating scales (e.g., one person's 3 is another person's 5).
3. **Stability:** Using multiple seeds and 5-fold cross-validation ensures the model works on new data, not just the data it was trained on.
4. **Decent Accuracy:** This method ended up with an accuracy score of 45.4

5 Experimental Results

The model was evaluated using a 5-fold cross-validation strategy. Each fold consists of 80,000 training samples and 20,000 test samples. The pipeline involves training multiple AutoRec seeds, optimizing the blend weight, and final ensemble classification.

5.1 Gradient Boosting Ensemble

Fold	**LightGBM**	**XGBoost**	**CatBoost**	**Ensemble Total**
Fold 1	44.80%	45.04%	45.00%	45.26%
Fold 2	45.71%	45.98%	45.69%	46.04%
Fold 3	45.36%	45.50%	45.59%	45.58%
Fold 4	45.28%	45.17%	45.29%	45.55%
Fold 5	44.58%	44.50%	44.46%	44.60%
Average	45.15%	45.24%	45.21%	**45.40%**